

Effectiveness of Kinesiology Taping on Pain, Range of Motion and Functional Performance in Athletes with Acute Ankle Sprain

Amarnath. S, Dr. Kushal Sharma

Abstract— Acute ankle sprain is one of the most prevalent musculoskeletal injuries in athletes, resulting in significant pain, reduced range of motion, and impaired functional performance. Kinesiology Taping (KT) has emerged as a widely used adjunct physiotherapy intervention; however, its comparative effectiveness against conventional management remains an area of active clinical investigation. To evaluate this, a randomized controlled trial was conducted with 60 athletes diagnosed with acute Grade I or II lateral ankle sprains. Participants were allocated into Group A (KT + conventional physiotherapy) and Group B (conventional physiotherapy alone). Pain was assessed using the Visual Analogue Scale (VAS), range of motion (ROM) using a goniometer, and functional performance via the Foot and Ankle Ability Measure (FAAM) at baseline, day 7, and day 14. Both groups demonstrated significant within-group improvement ($p < 0.05$). Inter-group analysis revealed that Group A showed statistically superior recovery in pain reduction, ankle dorsiflexion ROM, and FAAM-Sports scores compared to Group B ($p < 0.001$). Kinesiology Taping combined with conventional physiotherapy is safe, cost-effective, and highly efficient in accelerating sport-specific functional outcomes.

Index Terms— Ankle Sprain, Athletes, FAAM, Kinesiology Taping, Range of Motion, VAS.

I. INTRODUCTION

Ankle sprain represents a major public health concern in sports medicine, accounting for approximately 15–25% of all sports-related injuries. It constitutes the single most common injury across high-impact athletic disciplines such as football, basketball, and volleyball. The immediate physical consequences—pain, swelling, and functional incapacity—frequently lead to prolonged athletic absence and a high risk of developing chronic ankle instability (CAI). Studies indicate that athletes face an alarmingly high recurrence rate estimated at 40–70% over a competitive career due to inadequate rehabilitation or premature return to sport.

The acute management of ankle sprain has shifted from the traditional RICE protocol to the POLICE principle, which emphasizes early controlled loading to stimulate tissue repair. While structured exercise remains the gold standard, the competitive demands for accelerated recovery timelines place extensive pressure on modern rehabilitation protocols.

This has driven substantial interest in adjunct therapeutic modalities such as Kinesiology Taping (KT).

Developed by Dr. Kenzo Kase in the 1970s, KT is an elastic, latex-free cotton fabric tape designed to mimic the physical properties of human skin. Unlike rigid athletic tape, it allows full joint range of motion while providing continuous therapeutic stimulation. Proposed mechanisms include gate control pain modulation, enhanced proprioceptive feedback, and improved lymphatic drainage. Despite its widespread adoption, the scientific evidence base remains heterogeneous. This single-blind randomized controlled trial (RCT) was designed to evaluate the short-term effectiveness of KT combined with standard physiotherapy versus standard physiotherapy alone in an Indian athletic population.

II. METHODOLOGY

A. Study Design and Setting

The study was conducted as a prospective, single-blind, parallel-group Randomized Controlled Trial (RCT) at Dhakad Physio, Pratap Nagar, Jaipur. Sixty competitive athletes aged 18–35 years with clinically confirmed acute Grade I or II lateral ankle sprains within 72 hours of injury were enrolled.

Participants were randomly assigned via block randomization (ratio 1:1) to either Group A (Experimental: KT + Standard Physiotherapy, $n=30$) or Group B (Control: Standard Physiotherapy alone, $n=30$). The outcome assessor was strictly blinded to group allocation.

B. Inclusion and Exclusion Criteria

- **Inclusion Criteria:** College, district, or state-level athletes; age 18–35 years; acute Grade I or II sprain within 72 hours; and independent weight-bearing capacity.
- **Exclusion Criteria:** Grade III ankle sprains or fractures; contact dermatitis or allergy to KT adhesive; open wounds; and high ankle syndesmotism sprains.

C. Intervention Protocols

- **Group A (KT + Standard Physiotherapy):** Received standard physiotherapy paired with Kase's lateral ankle protocol using Kinesio Tex Gold™. The 5-strip technique included: (1) a zero-tension calcaneal anchor strip; (2) a Y-strip for peroneal facilitation with 25% tension; (3) an I-strip for direct ATFL support with 50–75% tension; (4) a figure-8 mechanical stabilization strip with 50% tension; and (5) a small zero-tension space

Amarnath. S, Post-Graduate Scholar, Department of Sports Physiotherapy, College of Physiotherapy, Apex University, Sitapura Campus, Jaipur, Rajasthan

Dr. Kushal Sharma, Assistant Professor, Research Cell & Department of Physiotherapy, College of Physiotherapy, Apex University, Jaipur, Rajasthan, India

Effectiveness of Kinesiology Taping on Pain, Range of Motion and Functional Performance in Athletes with Acute Ankle Sprain

correction strip directly over the point of maximum tenderness to facilitate decompressive pain relief. Tape was replaced every 72 hours across the 14-day intervention period.

- **Group B (Standard Physiotherapy Only):** Received an identical daily 45-minute progressive protocol:

- *Phase 1 (Days 1–3):* POLICE protocol (Protection, Optimal Loading, Cryotherapy, Compression, Elevation, and passive ROM circles).
- *Phase 2 (Days 3–7):* Active rehabilitation (active ROM, towel scrunches, calf stretches, and full weight-bearing).
- *Phase 3 (Days 7–14):* Progressive loading (Theraband resistance, single-leg stance balance training, wobble board stabilization, and walking-to-jogging progressions).

D. Outcome Measures

Assessments were performed at Baseline, Day 7, and Day 14. Pain was quantified using a 0–10 Visual Analog Scale (VAS) during walking. Sagittal plane active Range of Motion (dorsiflexion and plantarflexion) was calculated via a 12-inch metallic goniometer. Sport-specific physical function was self-reported using the validated Foot and Ankle Ability Measure (FAAM), split into Activities of Daily Living (ADL, maximum 84 points) and Sports (maximum 32 points) subscales.

III. RESULTS AND ANALYSIS

Statistical analysis was conducted using SPSS Version 26.0, with significance set at $p < 0.05$. Baseline demographic characteristics (including age, gender distribution, and body mass index) and sport-wise distribution showed no significant differences between the two groups ($p > 0.05$), confirming effective randomization.

Table I: Demographic and Baseline Data Comparison

Variable	Group A (n=30)	Group B (n=30)	p-value
Age (years)	22.4 ± 3.2	23.1 ± 3.5	0.42 (NS)
Gender (M:F)	20:10	19:11	0.78 (NS)
BMI (kg/m ²)	23.3 ± 2.1	23.2 ± 2.3	0.87 (NS)
Injury Grade (I:II)	14:16	15:15	0.80 (NS)

A. Pain Score Analysis (VAS)

Intragroup evaluation showed a highly significant drop in pain for both arms from baseline to Day 14 ($p < 0.001$). However, intergroup independent t-tests confirmed that the KT group (Group A) achieved significantly lower pain metrics compared to Group B at Day 7 and Day 14, yielding an exceptionally large clinical effect size (Cohen's $d = 1.41$ at Day 14).

Table II: Intergroup VAS Pain Score Progression

Time Point	Group A	Group B	Mean Diff.	p-value	Cohen's d
Baseline	7.2 ± 1.1	7.1 ± 1.3	0.1	0.75	0.08
Day 7	4.8 ± 1.0	5.9 ± 1.2	1.1	0.001	0.99
Day 14	2.6 ± 0.9	4.1 ± 1.2	1.5	<0.001	1.41

B. Range of Motion Analysis (ROM)

Group A achieved near-normal active dorsiflexion mobility by Day 14 ($16.8^\circ \pm 1.5^\circ$), whereas the conventional physiotherapy cohort remained below the critical structural functional threshold ($13.2^\circ \pm 1.7^\circ$). Plantarflexion restoration was similarly superior in Group A ($41.6^\circ \pm 2.4^\circ$) compared to Group B ($37.3^\circ \pm 2.8^\circ$).

Table III: Active Ankle Range of Motion Profiles

Variable / Time	Group A	Group B	Mean Diff.	p-value	Cohen's d
Dorsiflexion (°)					
Baseline	8.3 ± 2.1	8.5 ± 2.3	0.2	0.72	0.09
Day 7	12.6 ± 1.8	10.4 ± 1.9	2.2	0.001	1.19
Day 14	16.8° ± 1.5°	13.2° ± 1.7°	3.6	<0.001	2.25
Plantarflexion (°)					
Baseline	28.4 ± 3.2	28.6 ± 3.5	0.2	0.82	0.06
Day 7	35.2 ± 2.8	32.1 ± 3.0	3.1	0.001	1.07
Day 14	41.6 ± 2.4	37.3 ± 2.8	4.3	<0.001	1.65

C. Functional Performance (FAAM Indices)

The self-reported functional capacity across daily living parameters and athletic exercises revealed remarkable progress in the experimental wing. The FAAM-Sports subscale, which demands rapid neuromuscular control during sport-specific tasks (e.g., cutting, running, jumping), showcased the highest overall clinical variance, yielding an intergroup effect size of $d = 2.08$ at Day 14.

Table IV: Intergroup FAAM Functional Metrics

Subscale / Time	Group A	Group B	Mean Diff.	p-value	Cohen's d
FAAM-ADL (0–84)					
Baseline	38.4 ± 6.2	37.8 ± 6.5	0.6	0.72	0.09
Day 7	54.6 ± 5.8	46.3 ± 6.1	8.3	<0.001	1.41
Day 14	68.9 ± 4.9	57.4 ± 5.3	11.5	<0.001	2.29
FAAM-Sports (0–32)					
Baseline	11.2 ± 2.8	11.0 ± 2.9	0.2	>0.05	0.07
Day 14	25.8 ± 2.6	20.1 ± 2.9	5.7	<0.001	2.08

IV. DISCUSSION

The findings of this study demonstrate that integrating Kinesiology Taping into acute lateral ankle sprain management significantly enhances clinical and sport-specific recovery compared to traditional exercise protocols alone.

The extensive pain mitigation observed in Group A can be primarily interpreted through Melzack and Wall's Gate Control Theory. The mechanical traction exerted by the elastic tape application continuously stimulates large-diameter A β tactile mechanoreceptive afferents within the dermis. These inputs activate inhibitory interneurons in the substantia gelatinosa of the dorsal horn of the spinal cord, competitively blunting the transmission of nociceptive impulses traveling via slow C and A δ fibers.

Furthermore, the mechanical recoil effect creates visible skin convolutions that lift the epidermis away from underlying fascial layers. This localized decompression expands the interstitial space, lowering mechanical pressure on chemically sensitized free nerve endings and facilitating active microcirculation and lymphatic drainage of localized inflammatory metabolites.

This rapid pain reduction effectively counteracts pain-induced motor inhibition and protective muscle guarding. By blocking nociceptive feedback loops that normally limit range of motion, KT enables earlier safe mechanical loading and active joint excursion. This neurological change accounts for the accelerated restoration of active dorsiflexion (16.8° vs. 13.2°), which serves as a critical parameter for athletic tasks like landing, squatting, and deceleration.

The notable improvements highlighted by the FAAM-Sports metrics underscore the value of secondary cutaneous proprioceptive pathways. Acute trauma disrupts specialized mechanoreceptors within the ATFL, delaying peroneal activation times and compromising dynamic joint stability.

Applying KT with targeted longitudinal tension activates skin mechanoreceptors (such as Ruffini endings and Meissner's corpuscles) during movement. This creates a surrogate tactile feedback channel that supplements the damaged ligamentous pathways. This enhancement provides athletes with increased mechanical confidence and improved dynamic postural control during explosive sports maneuvers.

From a clinical and economic perspective in the Indian healthcare context, routine KT incorporation presents a highly cost-effective paradigm. At a minimal material cost of approximately ₹150–200 per application, it functions as an accessible tool for competitive sports centers to minimize

recovery timelines and optimize safe return-to-sport trajectories.

V. CONCLUSION

This randomized controlled trial establishes that Kinesiology Taping combined with conventional progressive physiotherapy is significantly more effective than conventional physiotherapy alone in reducing acute pain, restoring functional sagittal joint range of motion, and accelerating sport-specific operational status in athletes managing Grade I and II lateral ankle sprains ($p < 0.001$).

The null hypothesis is rejected. KT acts as an accessible, safe, and highly effective management tool that should be routinely incorporated into standard evidence-based athletic rehabilitation protocols.

Future research should focus on extending follow-up windows (e.g., 6–12 months) to systematically track long-term injury recurrence rates and evaluate dynamic peroneal muscle activation using localized surface electromyography (EMG).

REFERENCES

- [1] M. E. Aguilar-Ferrández, A. M. Castro-Sánchez, et al., "Effects of kinesio taping on pain, range of motion, and disability in patients with lateral ankle sprains," *Journal of Manipulative and Physiological Therapeutics*, vol. 39, no. 8, pp. 541–550, 2016.
- [2] T. Ahmed, M. Rashid, and F. Khan, "The impact of ankle kinesio taping on pain management in patients with acute ankle sprain: A prospective clinical study," *Journal of Sports Rehabilitation*, vol. 33, no. 4, pp. 112–119, 2024.
- [3] A. T. Al-Shareef, M. T. A. Omar, and A. H. M. Ibrahim, "Effect of Kinesio taping on pain and functional performance in ankle sprain injury," *Physical Therapy and Rehabilitation*, vol. 3, no. 1, pp. 1–10, 2016.
- [4] S. Alam, P. Sharma, and R. Verma, "Effectiveness of Kinesio taping versus Mulligan's mobilization with movement on NPRS, FAAM (ADL), and FAAM (Sports) in athletes with ankle sprain," *International Journal of Health Research*, vol. 14, no. 2, pp. 45–54, 2025.
- [5] G. Bota, A. Ionescu, and D. Popescu, "The effects of Kinesio tape on acute ankle sprain: A systematic review," *Sports Medicine — Open*, vol. 11, no. 1, pp. 23–31, 2025.
- [6] J. L. Botia Gonzalez and J. D. Coronell-Alean, "Kinesiology taping and its role in chronic ankle instability: A critical review," *Journal of Physiotherapy and Physical Rehabilitation*, vol. 8, no. 2, pp. 301–309, 2023.
- [7] C. Doherty, C. Bleakley, E. Delahunt, and S. Holden, "Treatment and prevention of acute and recurrent ankle sprain: An overview of systematic reviews with meta-analysis," *British Journal of Sports Medicine*, vol. 51, no. 2, pp. 113–125, 2017.
- [8] D. T. P. Fong, Y. Y. Chan, K. M. Mok, P. S. H. Yung, and K. M. Chan, "Understanding acute ankle ligamentous sprain injury in sports," *Sports Medicine, Arthroscopy, Rehabilitation, Therapy & Technology*, vol. 1, no. 1, p. 14, 2009.
- [9] P. A. Gribble, C. M. Bleakley, et al., "Evidence review for the 2016 international ankle consortium consensus statement on the prevalence, impact, and long-term consequences of lateral ankle

Effectiveness of Kinesiology Taping on Pain, Range of Motion and Functional Performance in Athletes with Acute Ankle Sprain

- sprains," *British Journal of Sports Medicine*, vol. 50, no. 24, pp. 1496–1505, 2016.
- [10] T. Halseth, J. W. McChesney, M. DeBeliso, R. Vaughn, and J. Lien, "The effects of Kinesio taping on proprioception at the ankle," *Journal of Sports Science and Medicine*, vol. 3, no. 1, pp. 1–7, 2004.
 - [11] K. Kase, J. Wallis, and T. Kase, *Clinical therapeutic applications of the Kinesio taping method*, 2nd ed. Tokyo: Ken Ikai Co. Ltd., 2003.
 - [12] G. M. M. J. Kerkhoffs, M. van den Bekerom, et al., "Diagnosis, treatment and prevention of ankle sprains: An evidence-based clinical guideline," *British Journal of Sports Medicine*, vol. 46, no. 12, pp. 854–860, 2012.
 - [13] B. J. Kim and D. Y. Kim, "Immediate effects of ankle balance taping with Kinesiology tape on the dynamic balance of young players with functional ankle instability," *Journal of Physical Therapy Science*, vol. 29, no. 11, pp. 1989–1992, 2017.
 - [14] E. C. W. Lim and M. G. X. Tay, "Kinesio taping in musculoskeletal pain and disability that lasts for more than 4 weeks: Is it time to peel off the tape and throw it out with the sweat? A systematic review with meta-analysis," *British Journal of Sports Medicine*, vol. 49, no. 24, pp. 1558–1566, 2015.
 - [15] A. Malik, R. Singh, and K. Sharma, "Effectiveness of Kinesiology Taping on pain and functional performance in acute ankle sprain: A randomized controlled trial," *Indian Journal of Sports Physiotherapy*, vol. 18, no. 1, pp. 22–30, 2025.
 - [16] R. L. Martin, J. J. Irrgang, R. G. Burdett, S. F. Conti, and J. M. Van Swearingen, "Evidence of validity for the Foot and Ankle Ability Measure (FAAM)," *Foot and Ankle International*, vol. 26, no. 11, pp. 968–983, 2005.
 - [17] K. L. Moore, A. F. Dalley, and A. M. R. Agur, *Clinically oriented anatomy*, 7th ed. Baltimore: Lippincott Williams & Wilkins, 2014.
 - [18] P. C. S. Parreira, L. C. M. Costa, et al., "Current evidence does not support the use of Kinesio taping in clinical practice: A systematic review," *Journal of Physiotherapy*, vol. 60, no. 1, pp. 31–39, 2014.
 - [19] M. P. J. van den Bekerom, P. A. A. Struijs, L. Blankevoort, L. Welling, C. N. van Dijk, and G. M. M. J. Kerkhoffs, "What is the evidence for rest, ice, compression, and elevation therapy in the treatment of ankle sprains in adults?" *Journal of Athletic Training*, vol. 47, no. 4, pp. 435–443, 2012.
 - [20] B. R. Waterman, B. D. Owens, S. Davey, M. A. Zacchilli, and P. J. Belmont, "The epidemiology of ankle sprains in the United States," *Journal of Bone and Joint Surgery*, vol. 92, no. 13, pp. 2279–2284, 2010.